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Sorptivity function and water repellency determined by miniaturized infiltration data for characterizing soil water capture under different agricultural managements

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ABSTRACT

Improving soil water capture in the current climatic context is essential for agricultural production. Water sorptivity (S) as function of soil water content, together with water repellency characterization are two key hydraulic properties describing soil water capture, but also remaining poorly understood. The objectives of this work were: i- to determine the S under different soil water contents (θ) under different agricultural management; ii- to determine the soil water repellency at different soil water contents using the Beerkan Estimation of Soil Transfer algorithm (BEST) adapted for water repellent soils (BEST-*WR*), evaluating its performance as compared to the standard method of water repellency index (RI). Miniaturized tension infiltration experiments (1 cm infiltration radius) were conducted on undisturbed soil samples across a range of moisture contents under two contrasting agricultural managements (bare fallow, BF; cover fallow, CF). From miniaturized infiltration data, Sorptivity was determined using two different approaches, through the Leeds-Harrison et al. method (S_{LH}) and using the BEST-*WR* algorithm (S_{BEST}), which accounts for water repellency via an exponential scaling factor (α_{WR}). Results showed that S was highest in dry conditions, remained relatively constant until approximately 0.25 m^3m^{-3}, and then decreased towards saturation. Regarding the effects of agricultural management, CF significantly increased S across the entire moisture range compared to BF, enhancing the soil's water capture capacity. Both treatments exhibited slight soil water repellency, but the inclusion of cover crops shifted the soil water content of maximum repellency toward higher values (0.34 m^3m^{-3} for CF vs. 0.25 m^3m^{-3} for BF), suggesting that management modifies the threshold of wettability rather than the overall degree of repellency. Furthermore, the parameter α_{WR} obtained from the BEST-*WR* algorithm proved to be a sensitive indicator for differentiating hydrophobic behavior, particularly in dry conditions. Our results show that the use of miniaturized infiltration experiments with the adapted methodology of Leeds-Harrison et al. could be used for the S(θ) function determination in a relatively simple way on undisturbed soil samples. The determination of this relationship can be useful to understand the complex soil water dynamics under unsaturated conditions, complementing the basic soil functions.

1. Introduction

Longer drought periods and more intense precipitation events are projected for the future, creating a more complex scenario for agricultural production (IPCC, 2023). In this context, water capture processes in dryland agroecosystems have become increasingly critical. Soil water sorptivity (S, $LT^{-1/2}$) is an important yet underexplored hydraulic property in agricultural soils that quantifies the soil's capacity to rapidly absorb water under capillary forces (Koorevaar et al., 1983). Although sorptivity can be used to assess a soil's ability to capture water, it

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remains poorly understood and is rarely included in soil hydraulic characterization. Most studies estimate S from the early stage of field infiltration measurements; however, these estimates may be biased because gravitational forces can influence the infiltration rate even at short times (Vandervaere et al., 2000). Using a different approach, Leeds-Harrison et al. (1994) developed a laboratory method to measure S in soil aggregates using miniaturized infiltration experiments based on steady-state data rather than transient flow. Villarreal et al. (2017) adapted this method for undisturbed soil cores, improving the accuracy of S estimates relative to the traditional approach, and showed that S is a useful indicator of soil structure and water dynamics under different agricultural managements.

Another important yet underexplored aspect of S is its dependence on soil water content (θ, L^3L^{-3}). As soil moisture increases, S typically decreases due to reduced capillary driving forces. However, the functional form of S(θ) remains poorly understood. Studies using different methods and approximations have reported both linear (Chapman, 1970; Sharma et al., 1980) and exponential relationships (Stewart et al., 2013). Using large horizontal columns packed with disturbed soil, Villarreal et al. (2019) found that S remained approximately constant over the 0–0.3 m^3m^{-3} range, depending on soil type, and then gradually decreased as saturation was approached. Although this approach allows estimation of S(θ) from a single infiltration run, its representativeness may be limited because it relies on disturbed soil and the wideness of the columns can influence on the gravity-free condition. This phenomenon remains challenging in soil physics and neglecting the S–θ relationship could lead to inaccurate descriptions of soil water dynamics.

A key factor modulating soil water capture is soil water repellency (Hallett, 2007), commonly defined as a reduced affinity of soil for water. When present, soil water repellency lowers capillary driving forces and can hinder infiltration (Lichner et al., 2018). Soil water repellency is often transient, diminishing or disappearing as the soil wets. In general, soil water repellency tends to be strongest under dry conditions and declines with increasing moisture up to a critical water content, above which the soil becomes effectively hydrophilic (Doerr et al., 2000). However, the relationship between soil water repellency and soil water content can be more complex. Some studies have reported maximum soil water repellency near the wilting point (Weber et al., 2021). Kawamoto et al. (2007) found that peak repellency coincided with drainage of water from intra-aggregate pores, highlighting the role of pore structure in shaping this relationship. Notably, these findings were obtained in strongly hydrophobic soils (e.g., volcanic ash and forest soils). In contrast, the relationship between soil water repellency and soil water content has been far less studied in agricultural soils under different managements. Goebel et al. (2004) showed that small changes in water potential can substantially affect wettability under agricultural use and suggested that the moisture content at which soil water repellency is maximal is not necessarily near the wilting point and may vary among soils.

Soil water repellency can also be investigated by analyzing the shape of cumulative infiltration curves (Pachepsky and Karahan, 2022) or by parameterizing these curves (Di Prima et al., 2021). Water-repellent soils display characteristic cumulative infiltration patterns—classified as Types E, K, and L according to Pachepsky and Karahan (2022). These curves show an initially convex segment that may increase monotonically, shift to an approximately linear trend, or transition to a concave shape without a substantial linear section (Pachepsky and Karahan, 2022). More recently, Di Prima et al. (2021) introduced an adaptation of the widely used Beerkan Estimation of Soil Transfer parameters (BEST) algorithm (Lassabatere et al., 2006) to describe infiltration in water-repellent soils, called BEST-W*R*. BEST-W*R* incorporates an empirical exponential scaling factor to capture the convex cumulative infiltration behavior under repellency, allowing estimation of S. Combining both approaches and using a miniaturized tension infiltrometer (1.5 mm infiltration radius), Villarreal et al. (2024) characterized the hydrophobic behavior of earthworm macropores relative to the surrounding soil matrix. They reported that BEST-W*R* performed well for water-repellent surfaces when coupled with the miniaturized infiltration method and found an association between the BEST-W*R* scaling factor and the observed water-repellent curve types, suggesting that this parameter may help differentiate water-repellent from wettable soils.

Under current climatic conditions, it is crucial to develop soil health–oriented farming systems that optimize soil hydrological functioning, particularly soil water capture. However, important knowledge gaps remain regarding the behavior of S(θ) and soil water repellency, in relationship to different agricultural management. We hypothesize that i- the S(θ) relationship is affected by agricultural management, as are soil water repellency and its dependence on soil water content; and ii- applying the BEST-W*R* algorithm to mini-infiltration measurements enables hydraulic characterization and quantification of soil water repellency across contrasting management systems. Accordingly, the objectives of this study were to: i- determine soil water sorptivity at different water contents under different agricultural managements; and ii- assess soil water repellency across a range of water contents using BEST-W*R*, evaluating its performance against standard methods.

2. Materials and methods

2.1. Site and experimental design

The experimental site is located near the town of Chascomús, Argentina (35°44′37.61″ south and 58°03′10.22″ west). The soil is a fine, illitic, thermic abruptic Argiudoll (Soil Survey Staff, 2014), Luvic Phaeozem (IUSS Working Group WRB, 2007) with a A horizon (0–0.3 m depth, 24.3, 42.9 and 32.8% of clay, silt and sand, respectively, 6.0 soil-water pH (1:2.5 relation) followed by clay-illuvial Bt (0.3–0.6 m depth, 40.5, 59.0 and 0.5% of clay, silt and sand, respectively, 6.2 soil-water pH (1:2.5 relation)). The climate is temperate with mean annual precipitation (1977–2024) of 1041 mm and the annual potential evapotranspiration is 929 mm (SIGA, 2024). The experiment is a completely randomized experimental design and was established in 2018, with two management systems as treatments (plots of 20 m wide and 87 m long for each treatment). The treatments are: i-conventional management, with high frequency of bare fallow during winter (BF); ii-alternative management with high frequency of winter cover cropping (CF) (barley (Hordeum vulgare L.) and vetch (Vicia sp. L.)). For both treatments, maize (Zea mays L.) and soybean (Glicine max L.) were sown as summer crops, under no-tillage management, alternately. Termination of cover crops was done mechanically by crushing. Detailed description of the experiment can be found in Polich et al. (2023). The detailed crop sequence for each treatment can be seen in Table S1. The bulk density and total porosity of the A horizon (determined on March 2024) were 1.12 g cm^{-3} and 0.52 m^3m^{-3}, respectively, for BF and 1.10 g cm^{-3} and 0.53 m^3m^{-3}, respectively, for CF.

Soil sampling was carried out in March 2024, after maize harvest. In each treatment, homogeneous and representative 10 × 10 m areas along the plots were selected. Within these areas, a total of 10 undisturbed soil samples were collected (5 cm height, 5 cm diam., 98 cm^3 volume) from the first 5 cm in order to determine S(θ) and soil water repellency.

2.2. Soil properties determination

2.2.1. Soil water sorptivity at different soil moisture contents

Infiltration measurements were conducted using a miniaturized tension infiltrometer composed of a tube connected to a water reservoir and terminated with a porous sponge (radius: 4 mm), ensuring reliable hydraulic contact with the soil sample. The reservoir was placed on an analytical balance with a sensitivity of 0.001 g, interfaced with a computer for continuous data acquisition. Prior to each test, air bubbles within the device were removed to avoid measuring artifacts. Soil samples were positioned on a scissor jack and carefully raised until

contact with the sponge was established. Each experiment lasted approximately three minutes, during which the balance recorded, at one-second intervals, the decrease in reservoir mass corresponding to the volume of water infiltrated into the soil by capillary action. Steady state was reached at early times (approximately between 90 and 120 s) due to the size of the infiltration surface (Leeds-Harrison et al., 1994; Hallett et al., 2001; Villarreal et al., 2017).

To obtain different soil water contents above air-dried conditions, 10 ml of water was added from the bottom in each soil sample and letting them equilibrate in a sealed recipient for 10 days at 4 °C for avoiding microbial activity or mycelium development, following a similar procedure to de Jonge et al. (1999). The gravimetric water content was measured by weighing the sample immediately before soil water repellency determination, and transformed to volumetric water content (θ, m^3m^{-3}) using the soil bulk density. For each sample, 5 values of θ were achieved between air-dried condition and saturation. In order to determine the effects of soil moisture on soil hydraulic variables, the achieved soil volumetric water contents were classified in dry (< 0.3 m^3m^{-3}) and wet (> 0.3 m^3m^{-3}), based on the previous study by Villarreal et al. (2019), which correspond to the soil volumetric water content around field capacity (-33 kPa soil water pressure head). At each θ value, infiltration experiments at 0 hPa water pressure head were carried out and S_{LH} value for each sample and θ value was calculated as:

$$S_{LH} = \sqrt{\frac{Qf}{4br}} \tag{1}$$

Where Q (L^3T^{-1}) is the infiltrated flow rates of water at steady state (derived from the mass loss of the balance, taken water density as 1 gcm^{-3}), r is the infiltration radius (1 cm), f (-) is the air-filled porosity taken as 0.3, and b (-) is a shape parameter taken as 0.55 (White and Sully, 1987). These values were selected based on previous study for similar soil (Villarreal et al., 2017).

Also, cumulative infiltration curves were analyzed using the adapted Beerkan Estimation of Soil Transfer parameters (BEST) for water-repellent soils (BEST-W*R*).

Infiltration is estimated as:

$$i_{WR}(t) = i(t)(1 - e^{-\alpha_{WR}t}) \tag{2}$$

Where i_{WR} (t) (LT^{-1}) is the scaled infiltration rate, i (t) ($L\ T^{-1}$) is the unscaled infiltration rate (*i.e.*, non-water-repellent), and ($1 - e^{-\alpha WRt}$) is the exponential scaling factor in which the empirical parameter α_{WR} (T^{-1}) reflects the attenuation of infiltration due to water repellency.

The BEST-W*R* method is based on the two-term explicit transient infiltration model (Haverkamp et al., 1994), modified with the replacement of hydraulic conductivity by a function of sorptivity and the experimental steady-state infiltration rate (Lassabatere et al., 2006). Eq. (2) results in an expression for infiltration into a water-repellent soil (Di Prima et al., 2021) as:

$$i_{WR}(t) = i(t)(1 - e^{-\alpha_{WR}t}) = \left\{ \frac{S}{2\sqrt{t}} + \left[A(1-B)S^2 + Bi_s^{\exp} \right] \right\} (1 - e^{-\alpha_{WR}t}) \tag{3}$$

where t is time, S is the sorptivity, i_s^{exp} is the experimental steady-state infiltration rate, B is a coefficient that can be set equal to 0.467 for most soils with dry initial conditions (Bagarello et al., 2014) and A (L^{-1}) is defined as:

$$A = \frac{\gamma}{r(\theta_s - \theta_i)} \tag{4}$$

where θ_s and θ_i are the saturated and initial volumetric soil water contents and γ is a shape parameter that was set to 0.75 (Haverkamp et al., 1994) for geometrical correction of the infiltration front shape. The cumulative infiltration (I_{WR}, L) is obtained by integrating Eq. (3) with respect to time (t) (Abou Najm et al., 2021):

$$I_{WR}(t) = S\sqrt{t} - \frac{S\sqrt{\pi}}{2\sqrt{\alpha_{WR}}} \operatorname{erf}\left(\sqrt{\alpha_{WR}t}\right) + \left[A(1-B)S^2 + Bi_s^{\exp} \right] t - \frac{\left[A(1-B)S^2 + Bi_s^{\exp} \right](1 - e^{-\alpha_{WR}t})}{\alpha_{WR}} \tag{5}$$

where erf is the error function between the lower and upper integration limits. The values of S and α_{WR} are obtained by fitting predicted to the observed cumulative infiltration based on minimizing the sum of squared errors. For a detailed description of BEST-W*R* see Di Prima et al. (2021). For fitting cumulative infiltration curves, the Excel VBA version for BEST-W*R* were used (https://bestsoilhydro.net/downloads/).

2.3. Soil water repellency

After water infiltration, in the same samples and at each θ value, ethanol infiltration was carried out for ethanol sorptivity determination (Se) using the same device and following the same procedure described previously for water sorptivity. Then, RI (-) was calculated for each sample according to Hallett et al. (2001):

$$RI = 1.95 \quad \frac{S_e}{S_{LH}} \tag{6}$$

Where 1.95 is a correction factor arising from water and ethanol viscosity and surface tension relationship at 20°C (Hallet and Young, 1999). Since pairwise ethanol and water sorptivity measurements may result in possible misestimating of RI due to soil heterogeneity (Alagna et al., 2017), all possible combination of 10 by 10 ethanol and water sorptivity measurements at each θ value were calculated (Pekárová et al. (2015). The integrated trapezoidal area under the RI-θ curve was used to determine the total degree of soil water repellency, (RI_{AREA}, $1/m^3m^{-3}$) and the value of θ where maximum RI value occurred (θ_{RImax}, m^3m^{-3}) (Weber et al., 2021).

2.4. Statistical analysis

Two-way ANOVA with management system (CF and BF) and soil moisture (Dry and Wet) as factors, were performed in order to determine main effects on S, RI and α_{WR}. One-way ANOVA with management system (BF and CF) as factor was carried out in order to determine the effect on RI_{AREA} and θ_{RImax}. A test for normality (Shapiro–Wilks test) was performed. For comparing the means Fisher's least significant difference (LSD) test (Sokal and Rohlf, 1995) was used. The relative cumulative frequency of S_{LH}, S_{BEST}, α_{WR} and RI of bare fallow and cover fallow was calculated. Significance was determined at $p = 0.05$. Statistical analyses were performed in STATISTICA software (Statsoft, 2004).

3. Results and discussion

3.1. Sorptivity at different soil volumetric water content

Mean values of S as function of θ are shown in Fig. 1. The S_{LH} values ranged between 0.032 and 0.099 cm $s^{-1/2}$, being in the same range as the S values reported in a similar soil (Villarreal et al., 2017, Villarreal et al., 2019, Polich et al., 2022). On the other hand, the S_{BEST} values ranged between 0.005 and 0.037 cm $s^{-1/2}$, showing significantly lower values as compared to S_{LH}. Despite the low relative errors on the cumulative infiltration fittings provided by BEST-W*R* algorithm (< 1%) (Fig. 2), a bias was observed with an underestimation of S_{BEST} values as compared to S_{LH} (Fig. 3). This discrepancy in the S values between the employed approaches could be related to a dimensional factor. BEST-W*R* algorithm is valid for 1-D vertical infiltration, while LH methodology is based on Wooding (1968) equation, developed for a 3-D infiltration process. Another explanation for the observed bias could be parameter compensation of the model or model-structure mismatch. However, both S_{LH} and S_{BEST} showed the same behavior respect to θ

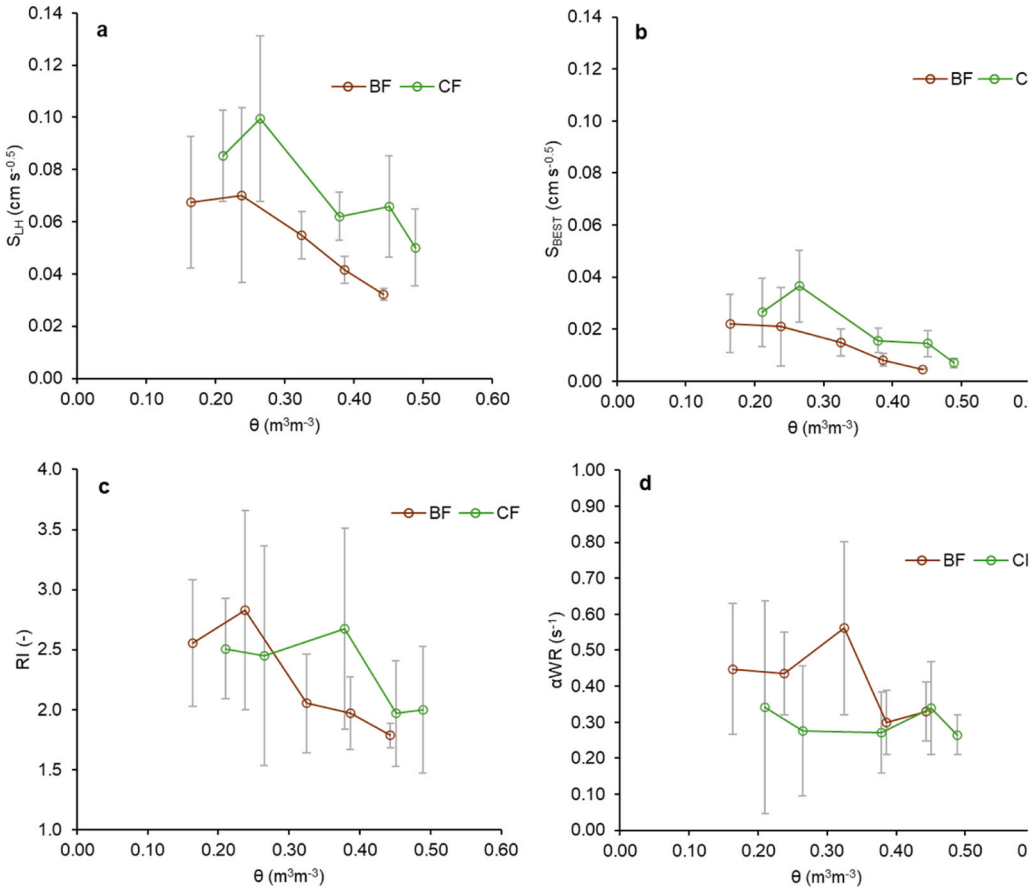


Fig. 1. Soil water sorptivity at different soil water content (θ), determined by Leeds-Harrison method (a, S_{LH}) and BEST algorithm (b, S_{BEST}), water repellency index (c, RI) as function of θ, and exponential scaling factor from BEST-W*R* algorithm (d, α_{WR}) for each agricultural management (bare fallow, BF; cover fallow, CF). Vertical bars denote standard deviation.

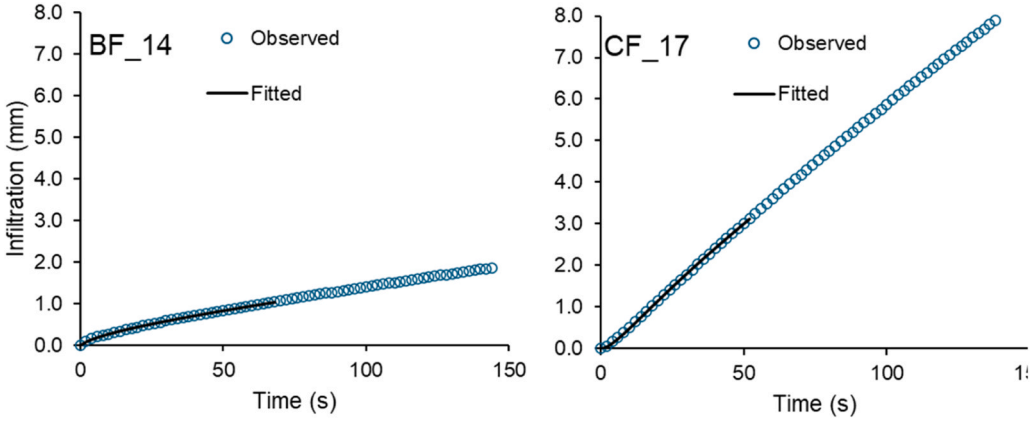


Fig. 2. Examples of observed and BEST-W*R* method predicted values of infiltration under cover fallow (CF) and BF (BF) treatments.

(Fig. 1a and b), showing that BEST-W*R* algorithm can be employed for the analysis of miniaturized infiltration data, with an accurate soil hydraulic characterization.

Higher values of S were observed at the lower θ, remaining practically constant to approximately 0.25 m^3m^{-3} and decreasing gradually towards saturation (Fig. 1a). The observed behavior agrees with Villarreal et al. (2019). They found in a similar soil from the same region constant values of S up to 0.3 m^3m^{-3}, followed by a linear decrease towards soil saturation. Our results also agree with different authors who reported similar S(θ) relationship (Stewart et al., 2013) but disagree with Sharma et al. (1980) who reported a linear S decrease across the whole saturation range. The dependence of S with the initial θ is an important parameter to consider when soil water capture is evaluated in agricultural soils. The gradual decrease in the wetting front potential as initial θ increases in loam soils with uniform pore sizes leads to a gradual decrease in capillary forces (Stewart et al., 2013; Villarreal et al., 2019), decreasing its water capture capacity. However, even when S depends on the initial soil water content, in general it is determined without mentioning the initial θ value when management effects are evaluated. Our results show that the initial soil moisture condition is an affecting

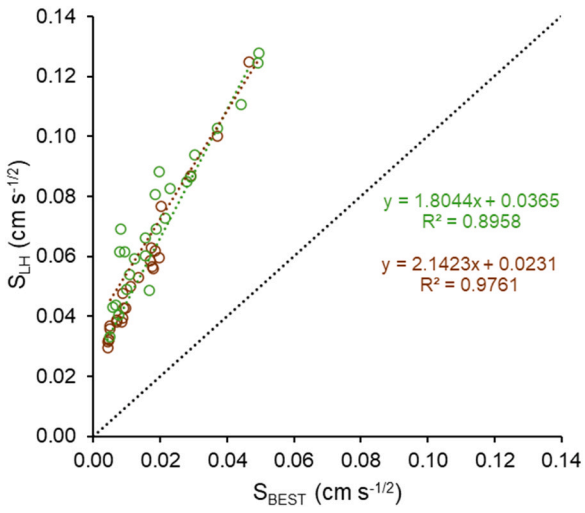


Fig. 3. Soil sorptivity values determined by the Leeds-Harrison method (SLH) against BEST-*WR* algorithm (SBEST) for each agricultural management (bare fallow, brown dots; cover fallow, green dots). Regression analysis results are shown. Dashed black lines represents 1:1 relationship.

factor for S determination and ignoring the S(θ) relationship could lead to results misinterpretation.

3.2. Effects of agricultural management on S and its relationship with water repellency

Regarding the effects of the two contrasting agricultural managements, CF treatment showed higher S_{LH} values at the two soil moisture conditions, dry and wet (Table 1), showing that more intense crop rotations improve soil water capture in the full saturation range. This agrees with previous studies which found an S increment under more intense crop sequences (Shaver et al., 2013) or after the inclusion of CC (Villarreal et al., 2022). Overall, in contrast to S, no differences between agricultural management were observed for RI (Table 2, Fig. 4). According to the obtained RI values, both treatments were slightly water repellent, making difficult to quantify soil water repellency behavior through RI (Bisdom et al., 1993; Smettem et al., 2021; Lichner et al., 2013). The RI behavior at different soil water contents showed higher RI values at dry moisture conditions as compared to wet conditions only under BF management (Table 2). Furthermore, no statistical differences between CF and BF were observed on RI_{AREA} (Table 2), showing that the degree of soil water repellency was not affected by agricultural management. However, under CF the θ_{RImax} shifted to higher values as compared to BF (Table 2), showing that the inclusion of cover crops can affect the threshold at which the soil can behave as water repellent, rather its degree. Previous studies reported similar observations; for example, in a pioneer work, de Jonge et al. (1999) reported that soil with perennial grass cover was more water repellent than the soil under barley cultivation, also showing higher values of θ_{RImax}. These authors attributed these differences to the different types of waxes between the two crops. Wijewardana et al. (2016) reported in soils from New Zealand under pasture that at higher SOC, soils tend to maintain their soil water repellency behavior toward higher θ values. Recently, Szatylowicz et al. (2024) found that soils under intensively used grassland showed higher values of θ_{RImax} as compared to similar soils but under birch forest.

Table 2
Mean values of integrated trapezoidal area (RI_{AREA}, $1/m^3m^{-3}$) and value of θ where maximum RI value occurred (θ_{RImax}, m^3m^{-3}) for each treatment (bare fallow, BF; cover fallow, CF).

Treatment	RI_{AREA}	θ_{RImax}
	$1/m^3m^{-3}$	m^3m^{-3}
BF	0.64 A	0.25 A
CF	0.67 A	0.34 A

Different capital letters denote statistical significative differences between treatments ($p < 0.05$).

On the other hand, the parameter α_{WR} showed higher values under BF as compared to CF (Table 2, Fig. 4). These results can indicate the potentiality of α_{WR} as indicator of soil water repellency in slightly water repellent soils. Di Prima et al. (2021) reported that the use of α_{WR} as a hydrological soil water repellency index allowed to describe better the soil water repellent behavior under oak trees canopies as compared to the open spaces. Similarly, Villarreal et al. (2024) found lower α_{WR} values in water-repellent infiltration curves, as compared to non-water-repellent, indicating that this parameter could be used for soil water repellency characterization. Although there were no differences between dry and wet conditions, the differences between treatments were more evident under the dry conditions (Fig. 1, d). In general, soil water repellency is more severe in dry soil and decline as moisture increases until a critical soil water content, above which soil becomes hydrophilic (Doerr et al., 2000).

Soil water repellency can be a result of the complex interaction between pore structure and hydrophobic substances (Li et al., 2021). In this sense, the expected increase of organic substances under CF can reduce soil wettability, but the change in pore structure increasing soil porosity and its connectivity can lead to infiltration and water sorptivity improvement. For these reasons, to understand which soil physical parameters affect the occurrence and severity of soil water repellency occurrence and severity needs to be further study (Doerr et al., 2006; Davis and Park, 2024).

3.3. Novelty and limitations of this study

Soil water sorptivity as function of θ is rarely determined although this relationship is well-known in the theory. The main contribution of our research is methodological, showing that the use of miniaturized infiltration experiments with the adapted methodology of Leeds-Harrison et al. (1994) could be used for the S(θ) function determination in a relatively simple way on undisturbed soil samples. To quantify this relationship can help to understand the complex soil water

Table 1
Soil sorptivity values determined by the Leeds-Harrison method for water (S_{LH}) and ethanol (Se), water repellency index (RI) and exponential scaling factor from BEST-*WR* algorithm (α_{WR}) for each agricultural management (bare fallow, BF; cover fallow, CF) and moisture condition (dry and wet).

Treatment	Moisture	S_{LH}			Se			RI			α_{WR}		
		cm $s^{-1/2}$			cm $s^{-1/2}$			-			s^{-1}		
BF	Dry	0.067	a	B	0.08	a	B	2.59	a	A	0.438	a	A
	Wet	0.042	b		0.04	b		1.96	b		0.396	a	
CF	Dry	0.092	a	A	0.11	a	A	2.48	a	A	0.312	a	B
	Wet	0.059	b		0.06	b		2.21	a		0.292	a	

Different capital letters denote statistical significative differences between treatments ($p < 0.05$) and different lowercase letters indicate differences between moisture conditions ($p < 0.05$).

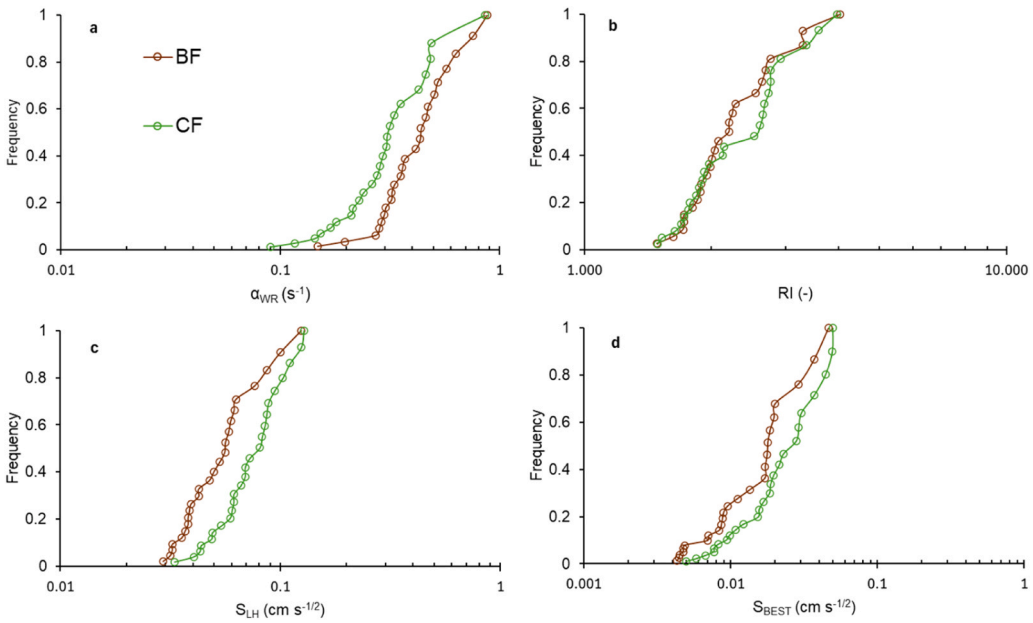


Fig. 4. Relative cumulative frequency distributions for α*WR* parameter (a), water repellency index (RI) (b), soil water sorptivity determined by Leeds-Harrison method (S_{LH}) (c) and by BEST-*WR* algorithm (S_{BEST}) (d) for bare fallow (BF) and cover fallow (CF) managements.

dynamics under unsaturated conditions, complementing the basic soil functions such as the soil water retention curve and the hydraulic conductivity curve. Moreover, the employed approach used in this work allowed to determine the effects of agricultural management on the S(θ), which is not common in soil physics studies. However, further studies should explore models for repeated measures in order to improve the applicability of the proposed methodology. To identify the soil ability to rapidly capture water in the whole saturation range becomes crucial in the current climatic scenario. In this sense, the inclusion of winter cover crops during fallow improves the water capture along the entire saturation range. The current trends of climate change predictions indicate an increase in evapotranspiration, with a greater impact of drought periods, together with higher intense precipitation events (IPCC, 2023). In this context, the use of cover crops could improves water saving.

However, it is important to mention that miniaturized infiltration is useful for detecting local surface sorptivity and water-repellency effects and its representation at field-scale should be carefully examinate.

Additionally, our results showed that BEST-*WR* algorithm can be applied to miniaturized infiltration data obtained in undisturbed soil samples for sorptivity determination. Moreover, this method allowed to characterize the water repellent behavior in slightly hydrophobic agricultural soils through the determination of α_{WR} parameter.

Since our results are based on a single soil type, further studies should evaluate the S(θ) function in different soil textural classes, which is a main affecting factor as was pointed out by Villarreal et al. (2019). Efforts will be made to improve the understanding of this crucial soil function, including a more detailed description and its parametrization and its relationship with soil structure parameters such as aggregate stability, pore size distribution, organic carbon fractions and contact angle. This information could improve the modelling frameworks for soil water dynamics prediction.

4. Conclusions

Miniaturized tension infiltrometer applied to undisturbed samples enabled a simple and robust determination of the sorptivity as function of soil water content. This approach can complement classical retention and conductivity curves, providing a practical tool for characterizing soil water capture capacity in agricultural systems.

Cover cropping management increased soil water sorptivity across the entire moisture range, shifting the threshold of maximum repellency toward higher water contents. This indicates that management practices modify the persistence of hydrophobicity rather than its degree, enhancing soil resilience to droughts and intense rainfall. The BEST-*WR* algorithm applied to miniaturized infiltration data allows the sorptivity determination at different initial soil water content, together with the soil water repellency characterization.

Parametrization of S(θ) in diverse contexts will strengthen modeling frameworks for soil water dynamics and guide management practices adapted to climate change.

CRediT authorship contribution statement

Martín Palazzolo: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Rafael Villarreal:** Writing – review & editing, Methodology, Data curation. **Nicolás Guillermo Polich:** Writing – review & editing, Methodology, Investigation, Data curation. **Guido Lautaro Bellora:** Writing – review & editing, Methodology. **María Paz Salazar:** Writing – review & editing, Methodology. **Luis Alberto Lozano:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Formal analysis, Conceptualization. **C. Germán Soracco:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.soilad.2026.100119.

Data availability

Data will be made available on request.

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